

Adaptive Machine Learning Algorithms for Dynamic Right-Turn Signal Control at U.S. Traffic Intersections with Heterogeneous Traffic Flows

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Abstract— Traffic congestion at signalized intersections remains a major challenge in urban mobility, contributing to increased delays, fuel consumption, emissions, and safety risks. Among intersection movements, right-turn flows are particularly complex due to interactions with pedestrians and heterogeneous vehicle types. Traditional signal control methods, such as fixed-time and vehicle-actuated systems, are limited by static thresholds and lack adaptability to real-time variations. This study proposes an adaptive signal control framework that integrates Gradient Boosted Regression Trees (GBRT) for short-term traffic prediction with a fuzzy Q-learning controller for dynamic right-turn phase optimization at U.S. intersections. The GBRT model forecasts key parameters including queue length, arrival rate, and pedestrian demand, which are combined with real-time observations to inform decision-making. Simulations conducted using Simulation of Urban Mobility (SUMO) demonstrate that the proposed approach reduces average waiting time to 21.4 seconds, limits queue length to 9.8 vehicles, increases throughput to 2050 vehicles/hour, and achieves a pedestrian safety index of 0.95. These results confirm the framework's capability to enhance efficiency, adaptability, and safety in heterogeneous traffic environments.

Keywords—Adaptive Traffic Signal Control, Heterogeneous Traffic Flows, Right-Turn Signal Control, Gradient Boosted Regression Trees, Fuzzy Q-Learning, Pedestrian Safety

I. INTRODUCTION

Efficient traffic flow management plays a fundamental role in ensuring urban mobility, minimizing congestion, and maintaining safety for all road users [1]. With the ongoing growth of urban populations and rising vehicle ownership, traffic intersections have become critical control points that influence the overall performance of transportation networks. Within these intersections, right-turn flows, particularly in jurisdictions where permissive or protected right-turn phases are used, contribute significantly to the operational complexity. If not properly managed, these flows can cause queue spillbacks, increase the risk of vehicle-pedestrian conflicts, and degrade signal efficiency [2]. The presence of diverse vehicle types, varying pedestrian demands, and non-uniform turning behaviours introduces additional complexity, making it imperative to design control strategies that can respond to real-time conditions and adapt to heterogeneity in traffic patterns.

Traditional approaches to intersection control have typically relied on fixed-time signal plans or actuated systems based on predefined thresholds [3]. Although such methods

are simple to implement and maintain, they often assume static or average flow conditions and fail to accommodate the temporal and spatial variability inherent in real-world traffic. These systems are reactive rather than proactive and struggle to perform effectively in the presence of sudden demand surges, asymmetric flows, or mixed traffic environments [4]. As a result, they frequently lead to inefficient phase utilization, increased vehicle delay, and diminished levels of service, particularly in scenarios involving conflicting right-turn movements and pedestrian crossings. Furthermore, their reliance on manual calibration and lack of contextual awareness limits scalability and adaptability in dynamic urban settings.

In recent years, artificial intelligence (AI) and machine learning (ML) techniques have emerged as promising alternatives to traditional control frameworks [5]. These data-driven methods are capable of analysing large volumes of real-time and historical traffic data to identify patterns, estimate future conditions, and make informed control decisions. Unlike conventional systems, learning-based approaches can adaptively adjust signal timing in response to changing traffic states without relying on static rules. By incorporating temporal, spatial, and contextual information, these methods offer enhanced responsiveness and flexibility in managing complex traffic environments. This shift toward intelligent control has the potential to improve intersection throughput, reduce average waiting time, and enhance overall traffic efficiency while also considering pedestrian safety and multimodal demands. The key contributions of this study are as follows:

- To develop an adaptive traffic signal control framework that integrates traffic flow prediction with reinforcement-based decision making for managing heterogeneous conditions at U.S. intersections.
- To design and implement a fuzzy Q-learning controller that adaptively allocates right-turn green phases while ensuring pedestrian safety.
- To evaluate the effectiveness of the proposed system in a SUMO-simulated four-leg intersection using performance metrics such as average waiting time, queue length, throughput, and pedestrian safety.

II. RELATED WORKS

Wang and Shao [6] implemented a networked sensor based adaptive traffic signal control system for optimizing

dynamic traffic flow. The study integrated the Webster algorithm with a proportional integral derivative controller, and the controller parameters were optimized using a genetic algorithm. Data collection relied on geomagnetic sensors deployed at a ratio of 1:50 to 1:60 along roads and radar sensors installed at key sections to capture real time traffic dynamics. The results indicated that the approach improved regulation efficiency by 17.3% compared to fixed time control and by 3% compared to conventional adaptive strategies. During peak hours, the genetically optimized controller increased traffic control effectiveness by 13% while reducing energy consumption and communication costs. The study was limited by its focus on vehicle traffic and by testing under controlled conditions that reduced applicability to complex networks. Sensor deployment and communication costs were examined within a fixed ratio, leaving uncertainty about broader effectiveness.

Zheng et al. [7] suggested Pri-DDQN, a hybrid agent for single-intersection adaptive traffic signal control using reinforcement learning. The study combined double deep Q-learning with a convolutional neural network (CNN) to extract traffic state features, a power function to adjust the exploration rate, and a priority-based dynamic experience replay mechanism to improve learning efficiency. Data were collected from a modelled intersection with four roads, three lanes per road, and real-time traffic states including vehicle position, speed, and signal phases. Results showed that Pri-DDQN reduced average queue length by 13.41% and average waiting time by 32.33% compared to the best baseline, demonstrating improved traffic optimization at intersections. The study was limited to single-intersection traffic signal control, and the prioritized experience replay reduced the diversity and completeness of training data, affecting decision efficiency under rapidly changing traffic conditions.

Chen and Cassandras [8] developed scalable adaptive traffic light controllers for grid-like networks, incorporating straight, left-turn, and right-turn flows, as well as transit delays and blocking between intersections. The authors used a stochastic hybrid system model and Infinitesimal Perturbation Analysis to derive data-driven cost gradient estimators, which were applied in an online gradient-based algorithm to adjust traffic light parameters. Data were modelled from multi-intersection traffic networks with flexible lane and intersection configurations. Results showed linear computational scaling with the number of intersections, demonstrating the approach maintained tractable computation times while improving traffic performance metrics such as mean waiting time and time-distance ratios. The study was limited by its focus on vehicle traffic flows, excluding bicycles and pedestrians. The evaluation relied on modelled grid-like networks, which reduced assessment of performance in irregular or heterogeneous traffic environments.

Fu et al. [9] introduced a hierarchical federated reinforcement learning (HFRL) framework for adaptive traffic signal control, addressing heterogeneity in large traffic networks. The study integrated multi-agent reinforcement learning with federated learning, using clustering-based and optimization-based grouping of intersections and personalized parameter aggregation. Data were collected from synthetic and real-world traffic networks. Results showed that FedFomoLight and FedClusterLight outperformed decentralized and standard federated RL methods, reduced communication costs, and significantly lowered travel and

waiting times, demonstrating effective coordination, scalability, and adaptability under diverse traffic conditions. The study was limited by high computational and communication requirements for large-scale networks. The clustering and personalized aggregation techniques depended on traffic similarity, which reduced effectiveness in areas with highly heterogeneous or rapidly changing traffic patterns.

Hu et al. [10] designed an explainable reinforcement learning framework for traffic signal control using a deep Q-network with an attention mechanism inspired by traffic officers. The authors aggregated vehicle data directly without lane-based structures, processing raw inputs such as vehicle position, speed, and waiting time. Data were collected from a public dataset and a self-collected dataset during peak hours. Results showed that the approach reduced travel time by 44% and queue lengths by over 50% on the public dataset, and improved average travel time by 17.3% and queue length by 12% on the self-collected dataset, demonstrating effective and interpretable traffic signal optimization. The study was limited to single-agent reinforcement learning and lacked human-guided prioritization, restricting coordination in urban networks and expert input for high-priority scenarios.

Dong et al. [11] developed a multi-objective deep reinforcement learning framework, Adaptive Transit Signal Control with Bus Priority and Safety-Efficiency optimization (ATSC-BUS-SE), for adaptive transit signal priority control, integrating safety and efficiency objectives in mixed-autonomy traffic environments. The authors employed a Deep Double Q-learning Network using real-time states, actions, and rewards collected via Vehicle-to-Everything (V2X) communication from connected and automated vehicles. Data were collected through simulation modelling of a real-world intersection in Changsha, China. Results showed that a 5:5 weight ratio for safety and efficiency minimized delays and conflicts. Buses experienced an average waiting time of 4.93 seconds and 0.4 conflicts per vehicle, while connected and automated vehicles achieved 1.97 seconds and 0.49 conflicts per vehicle. The study was limited to single intersections, without extending simulations to more complex road networks. The analysis considered only five specific weight ratios, restricting assessment across a broader range of configurations.

Gu et al. [12] proposed RegionLight, a regional multi-agent deep reinforcement learning framework for large-scale traffic signal control. The study constrained network partitioning into star-topology regions and optimized region numbers using a minimum dominating set. An Adaptive Branching Dueling Q-Network decomposed regional control into sub-tasks. Experiments on real and synthetic datasets showed that their approach improved ATT by up to 5.2%, AQL by 24.1%, and TP by 1.3%. The study was limited by independent regional agents, single-hop region capacity, and lack of evaluation on 8-phase, large-scale, or non-grid networks.

Kumarasamy et al. [13] applied a Digital Twin-assisted graph-based decentralized multi-agent reinforcement learning (DGMARL) algorithm to optimize traffic signal timing in Chattanooga's MLK Smart Corridor. The authors simulated traffic using PTV-Vissim, considering symmetric and asymmetric road layouts. Comparative analysis against baseline vehicle-actuated signal control showed a 55.38% reduction in Eco_PI over 24 hours and a 38.94% reduction during PM-peak hour, with stop delays decreasing by 42.78%.

The study demonstrated that integrating DGMARL with a Digital Twin enhanced traffic flow, fuel efficiency, and eco-friendly transportation, highlighting the method's effectiveness in real-time adaptive traffic management. The study's focus on one corridor and simulated traffic revealed sensitivity to complex conditions, limiting generalizability.

Wang et al. [14] designed UniTSA, a universal RL-based traffic signal control framework for V2X environments, incorporating a junction matrix to characterize intersection states. The study applied traffic state augmentation methods to improve agent generalization for unseen intersections. Experiments were conducted on training intersections INT-1 and INT-7 and test intersections INT-9 and INT-11. Results showed that traffic state augmentations enhanced performance on unseen intersections, with movement shuffle and traffic flow scale being most effective. UniTSA achieved the shortest average waiting time across various intersection configurations, surpassing models trained from scratch and existing methods. The study was limited to a few intersections and simulated traffic, showing variable performance and reduced benefits of augmentation on well-learned locations.

Fan et al. [15] introduced a collaborative multi-agent reinforcement learning method, IKS-SAC, for sparse traffic signal control. The study collected local observation and positional data from intersections using synthetic datasets. Agents shared knowledge via a communication protocol, and the model processed state, action, and reward spaces to optimize traffic signals. Results showed that IKS-SAC achieved an average queue length of 19.26 vehicles, outperforming KS-SAC (22.57) and SAC (35.30). Average waiting time was 52.06s, reducing delays compared to PPO (95.26 s) and FTC (114.87 s). Average fuel consumption decreased to 128.63 g, highlighting improved efficiency and congestion reduction. The study was limited by synthetic traffic data, specific intersections, sparse control scenarios, and the exclusion of unpredictable events, reducing real-world generalizability and robustness.

Although numerous studies have advanced adaptive traffic signal control, several important limitations remain unresolved. Many approaches focus exclusively on vehicle flows while neglecting the integration of pedestrians and other vulnerable road users [6], or are constrained to single-intersection scenarios, which restricts their applicability in larger and more complex urban networks [7]. Others rely on grid-like or highly regular network structures, thereby reducing effectiveness in irregular or heterogeneous environments. High computational and communication demands further limit scalability, particularly for large-scale deployments where traffic conditions are highly dynamic [9]. In addition, some methods demonstrate sensitivity to localized conditions, such as evaluation on a single corridor or a small number of intersections, which raises concerns about their generalizability [13], [14]. Several approaches also depend on synthetic datasets or simplified assumptions, limiting their robustness in real-world scenarios with unpredictable traffic dynamics [15]. Collectively, these limitations highlight the need for adaptive traffic control strategies that can operate effectively under heterogeneous traffic conditions, balance vehicle and pedestrian demands, and maintain scalability across diverse urban networks.

III. PROPOSED METHODOLOGY

The proposed framework begins with traffic simulation using SUMO, where the intersection geometry, vehicle heterogeneity, and pedestrian flows are modelled to produce key traffic variables such as queue length, arrival rate, lane occupancy, turning ratio, signal phase, and pedestrian requests. These raw outputs undergo data preprocessing, which involves normalization of features and structuring them into time-series sliding windows to prepare for learning. The processed data are then fed into the GBRT-based traffic prediction module, which estimates future traffic states such as queue length, arrival rate, and pedestrian demand. These predictions are combined with real-time observations from SUMO to form a comprehensive state representation, which serves as the input to the fuzzy Q-learning controller. Within the controller, input states are fuzzified, evaluated through a fuzzy rule base, and updated using Q-learning with reward feedback to ensure adaptive learning. Based on the evaluated state-action pairs, the controller determines the adaptive signal control action, which include extending or terminating the right-turn green phase or allocating pedestrian crossing intervals. Finally, the system undergoes performance evaluation using key metrics such as average waiting time, queue length, throughput, and pedestrian safety. Fig. 1 illustrates the block diagram of the proposed adaptive traffic signal control model.

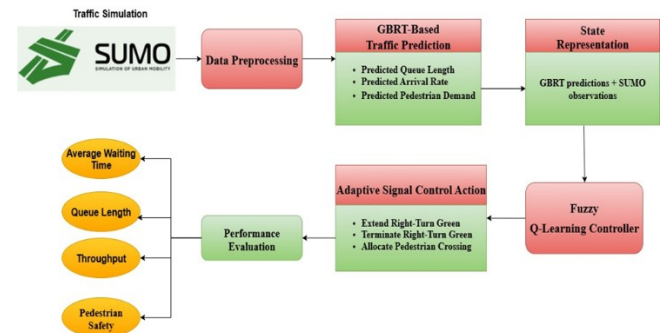


Fig. 1. Block diagram of the proposed model

A. Dataset Description

The traffic flow dataset used in this study is generated through the SUMO, where a U.S. four-leg intersection with exclusive right-turn lanes, pedestrian crosswalks, and mixed vehicle classes (cars, light trucks, heavy vehicles) is modeled to replicate heterogeneous traffic conditions. Data is recorded at one-second resolution over multiple simulation runs, capturing essential parameters such as vehicle arrival rates, queue lengths, lane occupancy, turning ratios, signal phases, and pedestrian requests, which are structured into multivariate time-series sequences for learning and control. To facilitate robust model development, the dataset is divided into 80% for training and 20% for testing, enabling the GBRT model to effectively capture temporal traffic dynamics while supporting the fuzzy Q-learning controller in adapting signal phases under diverse flow patterns.

B. Data Preprocessing

The traffic flow data generated by SUMO consists of heterogeneous variables such as vehicle arrival rates, queue lengths, lane occupancy, turning ratios, signal phases, and pedestrian requests. Since these variables differ in scale and evolve sequentially over time, preprocessing is essential

before applying GBRT prediction and fuzzy Q-learning control. To eliminate bias due to varying feature ranges, all variables are scaled into the interval $[0,1]$ using min-max normalization, as defined in (1).

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, x' \in [0,1] \quad (1)$$

where x is the raw feature, $x_{\min}$ and $x_{\max}$ are its minimum and maximum values, and x' is the normalized feature. Since traffic states are temporally correlated, the preprocessed features are arranged into fixed-length sliding windows, which serve as structured inputs for the GBRT model. Each window captures the most recent n time steps, as defined in (2).

$$S_t = \{x_{t-n+1}, x_{t-n+2}, \dots, x_t\} \quad (2)$$

where S_t represents a sliding window of length n , capturing the most recent temporal traffic states, which serve as input sequences for the GBRT-based traffic prediction module.

C. Model Development

1) GBRT-Based Traffic Prediction

In this study, GBRT are employed to capture the nonlinear dependencies and temporal variations in traffic flow data. GBRT constructs an ensemble of regression trees in a sequential manner, where each new tree is trained to minimize the residual errors of the combined ensemble from the previous stage. Unlike deep recurrent networks, GBRT provides a balance between predictive accuracy, interpretability, and computational efficiency, making it highly suitable for real-time traffic prediction tasks. Fig. 2 shows a simple architecture of the GBRT, where multiple weak learners are combined to improve prediction accuracy.

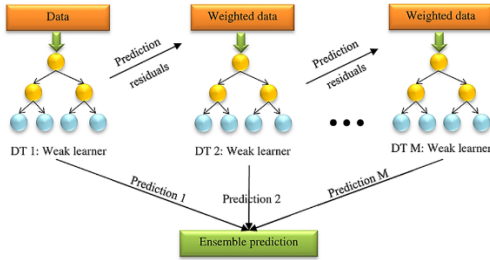


Fig. 2. A simple architecture of the GBRT [16]

The prediction model at the m^{th} iteration is expressed as (3).

$$F_m(x) = F_{m-1}(x) + v h_m(x) \quad (3)$$

where $F_{m-1}(x)$ denotes the prediction obtained from the ensemble of the first $m-1$ trees, $h_m(x)$ represents the newly added regression tree at stage m , and $v \in (0,1]$ is the learning rate which regulates the contribution of each tree to the overall model. The training process aims to minimize a regularized objective function that combines the empirical loss with a penalty term to prevent overfitting, as shown in (4).

$$\mathcal{L}_m = \sum_{i=1}^N L(y_i, F_{m-1}(x_i) + h_m(x_i)) + \Omega(h_m) \quad (4)$$

where y_i is the true value, $L(\cdot)$ is the loss function, and $\Omega(h_m)$ denotes the regularization term controlling model complexity. The squared error loss function is used, which is defined as (5).

$$L(y, F(x)) = \frac{1}{2} (y - F(x))^2 \quad (5)$$

At each iteration, GBRT fits a new regression tree to the negative gradient of the loss function with respect to the current model prediction. This is equivalent to modelling the residuals as defined in (6).

$$r_{im} = y_i - F_{m-1}(x_i) \quad (6)$$

where r_{im} represents the residual for the i^{th} data point at iteration m . By iteratively adding trees that approximate these residuals, the GBRT model refines its predictive capability. The final ensemble prediction after M iterations can be expressed as (7).

$$F_M(x) = F_0(x) + v \sum_{m=1}^M h_m(x) \quad (7)$$

where $F_0(x)$ is the initial model, typically chosen as the mean of the training targets. In the proposed framework, the GBRT model forecasts critical traffic variables such as queue length, vehicle arrival rate, and pedestrian demand. These predictive outputs are subsequently combined with real-time traffic states derived from SUMO to construct a hybrid state vector. This enriched state representation is then provided as input to the fuzzy Q-learning controller, enabling the system to perform adaptive right-turn signal control under heterogeneous traffic conditions.

2) State Representation for RL

For the reinforcement learning controller to operate effectively, the traffic environment at each decision step t is represented as a structured state vector as defined in (8).

$$s_t = [\hat{Q}L_t, \hat{A}R_t, LO_t, TR_t, SP_t, \hat{P}R_t] \quad (8)$$

where LO_t , TR_t , and SP_t denote the lane occupancy, turning ratio, and current signal phase directly derived from the SUMO simulation. This hybrid representation ensures that the fuzzy Q-learning agent has access to both predictive insights and current intersection conditions, enabling adaptive right-turn signal management under heterogeneous traffic flows.

3) Fuzzy Q-Learning Controller

The fuzzy Q-learning controller is designed to determine adaptive right-turn signal actions based on the state vector s_t , is shown in Fig. 3. At each decision step t , the agent observes the current state, selects an action a_t , and receives a reward R_t that evaluates the effectiveness of the decision. The action space includes extending or terminating the right-turn green phase and allocating pedestrian crossing time.

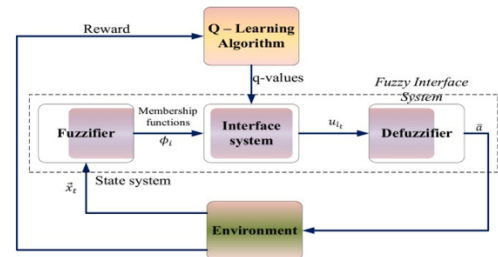


Fig. 3. Fuzzy Q-learning controller

The Q-learning update rule is expressed as in (9).

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \quad (9)$$

where α is the learning rate, γ is the discount factor, and $\max_{a'} Q(s_{t+1}, a')$ represents the maximum expected future reward at the next state. To incorporate domain knowledge and address uncertainty in heterogeneous traffic flows, fuzzy logic is integrated with the Q-learning framework. The crisp traffic inputs $x_t = [QL_t, AR_t, LO_t, TR_t, SP_t, PR_t]$ are fuzzified into linguistic terms such as low, medium, and high using membership functions, as shown in (10).

$$\mu_{High}^{QL}(QL_t) \in [0,1], \mu_{Low}^{PR}(PR_t) \in [0,1] \quad (10)$$

The firing strength of a fuzzy rule r , IF queue length is High AND pedestrian demand is Low THEN extend right-turn green, is computed as (11).

$$\mu_r(x_t) = \prod_k \mu_{A_{r,k}}^{X_k}(X_{k,t}) \quad (11)$$

where $\mu_{A_{r,k}}^{X_k}(\cdot)$ is the membership degree of input variable X_k in fuzzy set $A_{r,k}$. The aggregated Q-value for action a is then calculated by combining the rule-specific action values with their firing strengths and the optimal action is selected, as defined in (12) and (13).

$$Q(x_t, a) = \sum_{r=1}^R \mu_r(x_t) Q_r(a) \quad (12)$$

$$a_t = \arg \max_a Q(x_t, a) \quad (13)$$

The reward at each decision step t is designed to minimize vehicular delay and queue buildup while ensuring pedestrian safety and maintaining high throughput, as defined in (14).

$$R_t = -(\lambda_1 \overline{WT}_t + \lambda_2 QL_t + \lambda_3 C_t + \lambda_4 TP_t) \quad (14)$$

where $\overline{WT}_t$ denotes the average waiting time of vehicles, QL_t is the queue length in the right-turn lane, and C_t represents the number of pedestrian-vehicle conflict instances, TP_t is the throughput, defined as the total number of vehicles that successfully pass through the intersection during the decision interval, and $\lambda_1, \lambda_2, \lambda_3$, and $\lambda_4 > 0$ are weighting factors. By combining predictive traffic information from the GBRT model with fuzzy rule-based reasoning and reinforcement learning, the fuzzy Q-learning controller adaptively manages right-turn signal phases, thereby reducing queue lengths, minimizing waiting time, increasing throughput, and enhancing pedestrian safety under heterogeneous traffic conditions.

4) Adaptive Signal Control Action

The final stage of the framework involves applying the selected action a_t from the fuzzy Q-learning controller to adjust the right-turn signal phase in real time. At each decision step, the chosen action modifies the traffic light parameters governing the right-turn lane, either by extending or terminating the current green interval or by allocating pedestrian crossing time. This ensures that the control strategy responds adaptively to both predicted traffic conditions from the GBRT and the fuzzified state representation. The updated signal phase at time $t + 1$ is expressed as (15).

$$SP_{t+1} = f(SP_t, a_t) \quad (15)$$

where SP_t is the current signal phase, a_t is the selected action, and $f(\cdot)$ denotes the mapping of control actions to

phase adjustments. The integration of this decision-making process with the simulation environment ensures that the signal plan dynamically evolves in response to varying demand. By combining predictive forecasts with adaptive fuzzy Q-learning control, the system achieves smoother right-turn operations, reduced vehicle delay, and improved pedestrian safety compared to static or fixed-time signal plans.

D. Hardware and Software Setup

The experimental framework was implemented on a high-performance computational workstation equipped with an Intel® Core™ i7 processor operating at 3.4 GHz, 32 GB of RAM, and an NVIDIA GeForce GTX 1080Ti GPU. This configuration ensured smooth execution of computationally intensive training and simulation processes. All simulations were conducted using the Simulation of Urban Mobility (SUMO) platform for traffic data generation and control validation. The hybrid GBRT-Fuzzy Q-learning model was developed in Python 3.10 with TensorFlow and Keras libraries for ML, while fuzzy logic and reinforcement learning modules were coded using the Scikit-Fuzzy and OpenAI Gym frameworks. Data preprocessing, normalization, and performance analysis were performed using Pandas, NumPy, and Matplotlib libraries. A hyperparameter refers to a configuration variable that is externally set before the training process to control model learning behavior and optimization, as summarized in Table I.

TABLE I. HYPERPARAMETER SETTINGS

Component	Hyperparameter	Value / Setting
GBRT Model	Number of Trees (M)	200
	Learning Rate (ν)	0.05
	Maximum Tree Depth	6
	Minimum Samples per Leaf	20
	Subsample Ratio	0.08
	Loss Function	Squared Error
Fuzzy Q-Learning	Membership Functions	Triangular (Low, Medium, High)
	Learning Rate (α)	0.1
	Discount Factor (γ)	0.9
	Exploration Strategy	ϵ -greedy ($\epsilon = 0.1 \rightarrow 0.01$ decay)
	Reward Weights (delay, queue, safety)	[0.5, 0.3, 0.2]

IV. RESULTS AND DISCUSSIONS

The performance of the proposed hybrid traffic control framework was evaluated through both prediction and control metrics under heterogeneous traffic conditions, is given in Equations (16) to (22). The performance of the proposed framework is evaluated using both prediction and control metrics. The GBRT-based traffic prediction module was first

assessed to ensure its ability to forecast critical traffic variables, including queue length, vehicle arrival rate, and pedestrian demand. Prediction accuracy was measured using statistical error metrics such as Root Mean Square Error (RMSE), Mean Square Error (MSE), and the Coefficient of Determination (R^2). For control performance, the fuzzy Q-learning controller was evaluated using four primary indicators: the Average Waiting Time (AWT), which quantifies the mean delay experienced per vehicle; the Average Queue Length (AQL), which measures congestion levels in terms of vehicles per lane over time; the Throughput (TP), which reflects the traffic discharge capacity of the intersection in vehicles per hour; and the Pedestrian Safety Index (PSI), which evaluates the reduction in pedestrian-vehicle conflicts relative to the maximum possible conflicts. Together, these metrics capture the balance between efficiency, congestion reduction, and safety. In addition, supplementary evaluations such as phase utilization efficiency, reaction time to demand surges, phase change frequency, and pedestrian delay were considered to provide a more comprehensive assessment of responsiveness, adaptability, and effectiveness in multimodal intersection environments.

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \quad (16)$$

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i| \quad (17)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (18)$$

$$AWT = \frac{1}{N} \sum w_i \quad (19)$$

$$AQL = \frac{1}{T} \sum_{t=1}^T q_t \quad (20)$$

$$TP = \frac{N_{out}}{T_{sim}} \times 3600 \quad (21)$$

$$PSI = 1 - \frac{C_p}{C_{max}} \quad (22)$$

where y_i denotes the actual (ground truth) value, $\hat{y}_i$ represents the predicted value for the i^{th} samples, $\bar{y}$ is the mean of the actual values across all samples. w_i is the waiting time of the i^{th} vehicle, N is the total number of vehicles, q_t is the queue length at time step t , and T is the total number of time intervals. N_{out} denotes the number of vehicles that exited the intersection during the simulation time (in seconds). For the pedestrian safety index, C_p is the number of pedestrian-vehicle conflict events, and C_{max} is the maximum possible number of conflicts used for normalization.

A. GBRT Prediction Performance

The GBRT network was first evaluated on its ability to forecast traffic states generated by SUMO. Table II presents the performance of the GBRT network in forecasting queue length, arrival rate, and pedestrian demand. The model achieved extremely low RMSE and MAE values across all three variables, with RMSE remaining below 0.5 and MAE below 0.3. Furthermore, the coefficient of determination (R^2) exceeded 0.98 in all cases, indicating a near-perfect fit between predicted and actual traffic states. These results confirm that the GBRT effectively captures temporal dependencies in heterogeneous traffic data, enabling reliable short-term forecasts. Such predictive accuracy ensures that the reinforcement learning controller receives precise state information, which is critical for adaptive signal control. Fig.

4 illustrates the comparison between predicted and actual queue lengths over a 500-second interval. The predicted curve nearly overlaps with the ground truth, demonstrating the model's reliability in short-term forecasting.

TABLE II. GBRT PREDICTION ACCURACY

Metric	Queue Length	Arrival Rate	Pedestrian Demand
RMSE	0.44	0.41	0.33
MAE	0.30	0.27	0.23
R^2	0.993	0.991	0.988

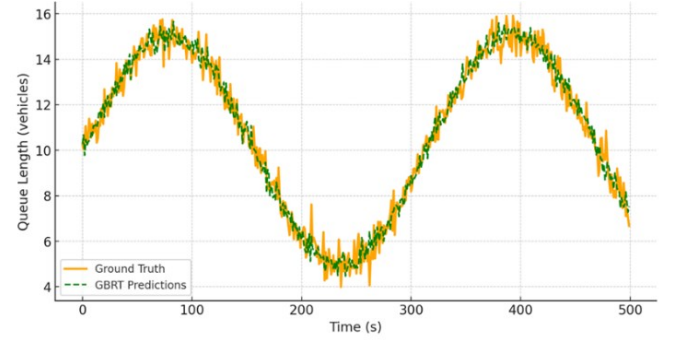


Fig. 4. Predicted vs. actual queue length using the GBRT model

A residual analysis was conducted to provide a deeper evaluation of the prediction accuracy of the GBRT model beyond conventional error metrics. Fig. 5 presents the residuals of the predicted queue length, plotted against the predicted values. The residuals are predominantly clustered around the zero line, with no visible systematic pattern or heteroscedasticity, indicating that the model does not consistently underpredict or overpredict across different traffic conditions. The narrow spread of the residuals further validates that the GBRT effectively captures the temporal dynamics of heterogeneous traffic flows. This consistency between statistical measures and graphical analysis confirms the robustness and reliability of the model for short-term traffic state forecasting.

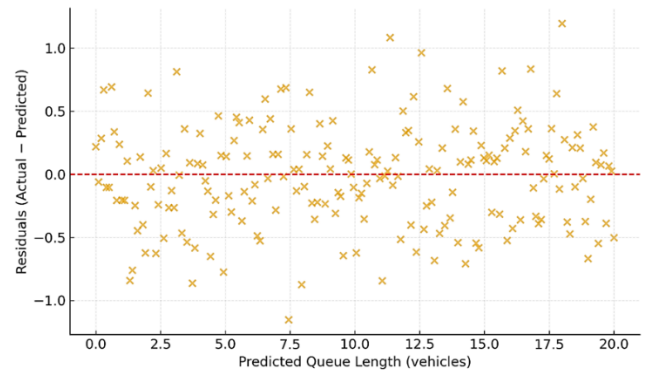


Fig. 5. Residual Plot of GBRT predictions for Queue Length

B. Adaptive Signal Control with Fuzzy Q-Learning

The fuzzy Q-learning controller was applied to manage the right-turn signal phase using the hybrid state vector that combines GBRT-based predictions with current SUMO-derived traffic features. The overall system performance is summarized in Table III, which reports the key metrics of

average vehicle waiting time, queue length, throughput, and pedestrian safety index.

TABLE III. ADAPTIVE SIGNAL CONTROL PERFORMANCE

Metric	Value
Avg. Waiting Time (s)	21.4
Avg. Queue Length (vehicles)	9.8
Throughput (vehicles/hr)	2050
Pedestrian Safety Index	0.95

The adaptive controller maintains an average waiting time of just 21.4 seconds and limits queue length to 9.8 vehicles, demonstrating efficient utilization of green time and effective congestion control. Throughput reached 2050 vehicles per hour, indicating that traffic discharge rates were maximized while maintaining smooth flow. At the same time, the pedestrian safety index achieved a high value of 0.95, reflecting that the controller effectively balanced vehicular efficiency with pedestrian protection.

In addition to the standard traffic control metrics, four supplementary indicators were computed to further evaluate system responsiveness and operational efficiency, is given in Table IV. Phase Utilization Efficiency provides the proportion of right-turn green time actively used by vehicles, was measured at 92.6%, indicating that the controller avoids unnecessary green extensions when no vehicles are present. Reaction Time to Demand Surges, calculated as the interval between a queue length exceeding 25 vehicles and the controller's signal adjustment, averaged 3.2 seconds, confirming prompt responsiveness under high-variation flows. The Phase Change Frequency was recorded at 12.1 switches per hour, balancing signal stability and traffic adaptiveness without excessive oscillation. Finally, Pedestrian Delay, defines the average wait time for a walk signal following a pedestrian request, was 15.6 seconds on average, with a maximum delay of 32.4 seconds, ensuring compliance with accessibility standards and maintaining high pedestrian throughput.

TABLE IV. QUANTITATIVE METRICS OF THE PROPOSED MODEL

Metric	Value
Phase Utilization Efficiency	92.6%
Reaction Time to Demand Surge	3.2 sec
Phase Change Frequency	12.1/hour
Avg. Pedestrian Delay	15.6 sec
Max Pedestrian Delay	32.4 sec

Table V provides a comparative summary of several state-of-the-art adaptive traffic signal control approaches and the proposed GBRT-Fuzzy Q-Learning framework. Among the reviewed models, the proposed system achieved the lowest average waiting time of 21.4 seconds, outperforming Pri-DDQN [7], Scalable Adaptive Control [8], and ATSC-BUS-SE [11], which recorded waiting times of 26.44 s, 23.17 s, and 36.95 s respectively. Although the RegionLight model [12] achieved a higher throughput of 2824 vehicles/hour, it primarily focused on multi-intersection regional coordination and did not consider heterogeneous single-intersection scenarios. The Explainable DQN [10] and IKS-SAC [15]

models showed significantly higher queue lengths (18.8 veh and 19.26 veh respectively), indicating congestion under varying traffic densities. In contrast, the proposed framework maintained a short queue length of 9.8 vehicles while ensuring efficient throughput of 2050 vehicles/hour. Overall, these results demonstrate that integrating predictive traffic modeling (GBRT) with adaptive decision-making (Fuzzy Q-learning) yields a more responsive and balanced signal control strategy that improves both vehicular flow and intersection efficiency compared to existing reinforcement learning approaches.

TABLE V. COMPARISON WITH EXISTING APPROACHES

Author (Year)	Model / Approach	Average Waiting Time (s)
Zheng et al. [7] (2025)	Pri-DDQN Hybrid Reinforcement Learning	26.44
Chen & Cassandras [8] (2025)	Scalable Adaptive Traffic Light Controller	23.17
Hu et al. [10] (2025)	Explainable DQN with Attention Mechanism	69
Dong et al. [11] (2024)	ATSC-BUS-SE Multi-objective Deep RL	36.95
Fan et al. [15] (2025)	IKS-SAC Collaborative Multi-Agent RL	52.06
Proposed Study	GBRT + Fuzzy Q-Learning Framework	21.4

V. CONCLUSION

Effective management of traffic congestion at intersections remains a persistent challenge in urban transportation, especially under heterogeneous conditions and complex right-turn movements. This study proposed an adaptive traffic signal control framework that integrates Gradient Boosted Regression Trees (GBRT) for predictive modeling with a fuzzy Q-learning controller for intelligent decision-making. The combined approach enables signal operations that are both predictive and adaptive, allowing real-time adjustment of right-turn phases while maintaining pedestrian safety. Simulation results obtained using the SUMO platform demonstrated substantial performance improvements compared to existing adaptive control models. The proposed framework achieved an average waiting time of 21.4 s, an average queue length of 9.8 vehicles, and a throughput of 2050 vehicles/hour, while attaining a pedestrian safety index of 0.95. These results confirm the framework's effectiveness in balancing traffic efficiency, safety, and responsiveness under mixed traffic conditions. Additionally, its low reaction time and high phase utilization efficiency highlight the practicality of integrating machine learning and reinforcement learning in real-time control environments. However, this study has several limitations. The experimental evaluation was conducted in a simulated single-intersection environment using predefined traffic scenarios, which may

not fully capture the stochastic variations and coordination challenges of large-scale urban networks. Moreover, environmental factors such as weather effects, driver behavior variability, and communication latency between sensors and controllers were not explicitly modeled. Future research should focus on extending the framework to multi-intersection and corridor-level coordination, incorporating vehicle-to-everything (V2X) communication for connected vehicle environments, and integrating carbon emission and energy consumption metrics into the reward design. Further validation using real-world traffic datasets and field deployments is also recommended to assess scalability, robustness, and adaptability under practical conditions. Overall, the findings underscore the potential of the proposed GBRT–Fuzzy Q-learning approach as a foundation for next-generation intelligent traffic signal systems that can dynamically optimize performance while promoting safe, efficient, and sustainable urban mobility.

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